SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Sprintly - Web-Based Kanban Project Management System

Version: 1.2

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1. Introduction

1.1 Purpose

This document provides a detailed **Software Requirements Specification (SRS)** for **Sprintly**, a web-based Kanban project management system.

It defines the **functional requirements, non-functional requirements, system architecture constraints, user characteristics, and overall system behavior**.

This document serves as a **formal agreement between stakeholders, developers, and evaluators** regarding the expected functionality and performance of the Sprintly system.

1.2 Intended Audience

This document is intended for:

- Project developers and software engineers responsible for implementation
- System architects and designers
- Project supervisors and academic evaluators
- Stakeholders and product owners
- Quality assurance and testing personnel

1.3 Project Scope

Sprintly is a **web-based Kanban project management system** designed for:

- Small teams
- Academic collaborations
- Early-stage startups
- The system provides the following core capabilities:
- Visual **Kanban board** with customizable workflow stages (e.g., *To Do*, *In Progress*, *Done*)
- **Role-based access control** to regulate user permissions
- **Mandatory task ownership assignment** to ensure accountability
- **Task movement across workflow stages using drag-and-drop**
- **Dashboard-based project overview** with task distribution visibility
- **Deadline tracking for tasks**

Sprintly aims to **enforce structured workflow management, increase accountability, and reduce task ambiguity** within collaborative environments.

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2. Overall Description

2.1 Product Perspective

Sprintly is a **standalone web-based application** designed to manage and track tasks efficiently.

The system follows a **three-tier architecture**.

1. Presentation Layer (Frontend)

- Framework: **React.js (v18.2.0)**
- Language: **TypeScript (v5.3)**
- Styling: **TailwindCSS (v3.4)**

Responsible for:

- Rendering user interface
- Client-side state management
- Interacting with backend APIs

Communication with the backend occurs via **RESTful APIs over HTTPS**.

2. Application Layer (Backend)

- Framework: **Spring Boot (v3.2.3)**
- Language: **Java (JDK 17 LTS)**
- Security: **Spring Security (v6.2)**
- Authentication: **Email/Password authentication with JWT-based stateless session management**

Responsibilities:

- Business logic processing
- Authentication and authorization
- API request handling
- Role-based access control enforcement

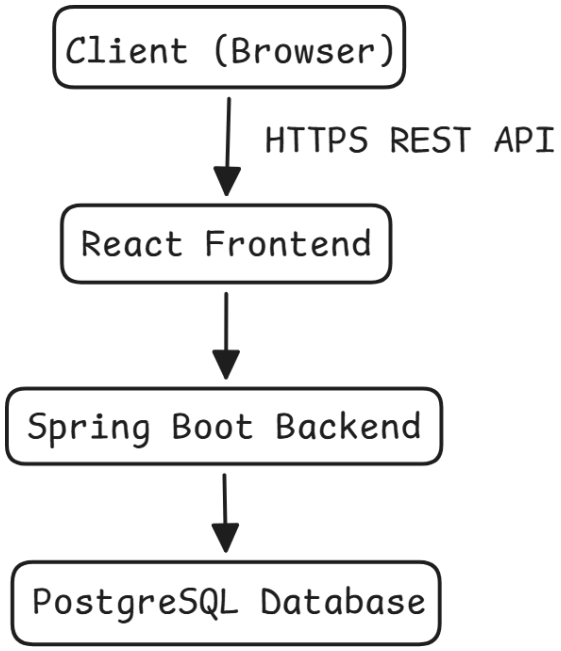
3. Data Layer (Database)

- Database System: **PostgreSQL (v15)**
- ORM: **Spring Data JPA (Hibernate v6.4)**

Responsibilities:

- Persistent data storage
- Query processing
- Data integrity and transaction management

System Architecture Overview



The system is **modular and scalable**, allowing future integration with external services.

All software dependencies shall use the latest stable patch version to ensure security updates and bug fixes.

2.2 User Classes and Characteristics

Sprintly supports **three user roles**, assigned at the **project level**.

Owner

Permissions:

- Create and manage projects
- Invite users to projects
- Create, assign, edit, and delete tasks
- Move any task across workflow stages
- Manage team roles and permissions

Characteristics:

- Highest level of project permissions
- Moderate familiarity with project management tools expected

Participant

Permissions:

- View project tasks and boards
- Move **tasks assigned to them**
- Comment on tasks
- View dashboard analytics

Characteristics:

- Limited modification privileges
- Expected to have basic web application usage skills

Viewer

Permissions:

- View tasks and project boards
- View project analytics and dashboard

Restrictions:

- Cannot create tasks
- Cannot modify tasks
- Cannot move tasks

Characteristics:

- Read-only access
- Minimal technical knowledge required

2.3 Operating Environment

Sprintly operates in the following environment.

Client-Side Environment

Supported Browsers:

- Google Chrome (latest 2 stable versions)
- Mozilla Firefox (latest 2 stable versions)
- Microsoft Edge (latest 2 stable versions)

Requirements:

- JavaScript enabled
- Cookies enabled
- Local Storage enabled

Recommended screen resolution:

1280 × 720 pixels or higher

Server-Side Environment

- Operating System: **Linux (Ubuntu Server 22.04 LTS)**
- Runtime: **Java JDK 17**
- Backend Framework: **Spring Boot 3.2**
- Database: **PostgreSQL 15**
- Web Server: **Nginx 1.24 (Reverse Proxy)**
- Deployment: **Docker (v25.0) optional**

All communication uses **HTTPS (TLS 1.2+)**.

Network Requirements

- Stable internet connection
- Minimum recommended bandwidth: **~1 Mbps**

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3. System Features

3.1 Functional Requirements

Authentication Requirements

FR-1: The system shall allow users to **register using a unique email address and password**.

FR-2: The system shall **validate user credentials before granting access**.

FR-3: The system shall **maintain authenticated sessions using JWT tokens**.

FR-4: The system shall allow users to **log out and invalidate active sessions**.

FR-5: The system shall support **password reset using email verification**.

Project Management Requirements

FR-6: Owners shall be able to **create new projects**.

FR-7: Owners shall be able to **invite users to projects via email invitation**.

FR-8: The system shall assign roles (**Owner, Participant, Viewer**) to project members.

FR-9: The system shall enforce **role-based access control for all project operations**.

FR-10: Owners shall be able to **remove members from a project**.

Task Management Requirements

FR-11: Each task shall have **exactly one assigned owner (Participant or Owner)**.

FR-12: Owners shall be able to **create, edit, assign, and delete tasks**.

FR-13: Participants shall be able to **update and move tasks assigned to them**.

FR-14: Owners shall be able to **move any task across workflow stages**.

FR-15: Viewers shall have **read-only access to tasks**.

FR-16: The system shall support **drag-and-drop task movement between workflow stages**.

FR-17: The system shall **record task status changes and timestamps**.

FR-18: Tasks shall support **deadline assignment and priority levels**.

Dashboard Requirements

FR-19: The system shall display the **number of tasks in each workflow stage**.

FR-20: The system shall calculate and display **project completion percentage**.

FR-21: The system shall display **tasks that have exceeded their deadline**.

FR-22: The dashboard shall display **tasks assigned to the current user**.

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4. Other Nonfunctional Requirements

4.1 Performance Requirements

- The system shall load **primary pages within 3 seconds** under normal load.
- The system shall support **up to 100 concurrent users**.
- API response time for standard operations shall **not exceed 1 second**.
- Task updates shall appear **within 5 seconds** through polling or refresh.

4.2 Scalability Requirements

- The system shall support **100 concurrent active users** in the initial deployment.
- The database shall support **at least 10,000 tasks per project**.
- The architecture shall support **horizontal scaling via container deployment**.

4.3 Availability Requirements

- The system shall maintain **minimum 99% uptime**.
- Scheduled maintenance shall occur during **low-traffic hours**.

4.4 Security Requirements

- All communication shall use **HTTPS encryption (TLS 1.2 or higher)**.
- Passwords shall be stored using **bcrypt hashing with salt**.
- Email/password authentication with **JWT-based stateless session management**
- The system shall enforce **role-based access control (RBAC)**.
- The system shall implement **input validation to prevent SQL injection and XSS attacks**.

4.5 Software Quality Attributes

Usability

The system shall provide a **clean and intuitive UI suitable for students and small teams**.

Maintainability

The backend architecture shall be **modular and layered to allow easy maintenance**.

Reliability

The system shall ensure **transactional data consistency during task updates**.

Responsiveness

The user interface shall be **responsive across desktop and tablet devices**.

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5. Other Requirements

5.1 Assumptions

- Users have **stable internet connectivity**.
- Users are familiar with **basic web application usage**.
- The hosting environment remains **properly configured and operational**.
- Users access the system using **modern web browsers**.
- External enterprise integrations are **not required in Version 1.2**.

5.2 Future Enhancements

Future versions of Sprintly may include:

- **WebSocket-based real-time task synchronization**
- **Advanced analytics and productivity tracking**
- **Workload distribution visualization**
- **AI-based task recommendations**
- **Native mobile applications (Android / iOS)**
- **Migration to scalable cloud infrastructure (AWS / Kubernetes)**